Institution Details

Province	Sindh	**City**	Karachi
Institution	National University of Computer and Emerging Sciences (FAST-NU)	**Campus**	Karachi
Department	Computer Science	**Degree Level**	BS
Degree Program	Computer Science	**Telephone**	
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Project Details

Project Title	Predicting Multicellular Function through Multi-layer Tissue Networks		
Group Details	**Member 1 Name:** Alishba Khan **Member 1 Roll#:** 21K-4505	**Member 2 Name:** Nabeeha Maqsood **Member 2 Roll#:** 21K-3384	**Member 3 Name:** Ashhad Abid **Member 3 Roll#:** 21K-3429
Project Area of Specialization	Deep Learning,Machine Learning, Artificial Intelligence		
Project Start Date	January 2025	**Project End Date**	December 2025
Project Summary (less than 2500 characters)	The project "Predicting Multicellular Function through Multi-layer Tissue Networks" aims to enhance our understanding of tissue-specific protein interactions using deep learning techniques. By leveraging multi-layer networks, we will develop a hierarchy-aware unsupervised learning algorithm to predict protein functions across different human tissues. Our approach builds on the OhmNet framework, which enables better insights into cellular functions, ultimately contributing to improved disease diagnosis and treatment strategies. **Key Objectives:** 1. Develop a hierarchy-aware unsupervised learning algorithm for predicting protein functions. 2. Utilize multi-layer networks to tissue-specific interactions. **Results**: 1) The OhmNet framework will be applied to a multi-layer protein interaction network comprising 107 human tissues. 2) OhmNet has demonstrated performance and methods in predicting cellular functions for 48 tissues with known tissue-specific functions. Overall, this project aims to provide a model for investigating tissue-specific protein functions and multicellular behaviors, ultimately enhancing our understanding of disease mechanisms.		
Project	The objective of this paper is to develop a multiscale computational model that addresses the limitations of existing network-based methods for studying protein functions across human tissues. Specifically, the paper aims to:		

Objectives (less than 2500 characters)	**1- Capture Tissue-Specific Protein Functions:** The paper highlights that existing methods assume cellular functions are constant across tissues, which is biologically inaccurate. Proteins can have different roles in different tissues, and current methods fail to predict these tissue-specific functions. **2- Model Multiscale Tissue Hierarchies:** The paper points out that current methods do not capture the multiscale organization of tissues, organs, and cellular systems. This limitation prevents the representation of complex biological patterns that span multiple levels of tissue organization. **3- Tissue-Specific Information:** The paper notes that while some methods use tissue-specific information, they primarily focus on improving predictions of global cellular functions or gene-disease associations, rather than predicting tissue-specific gene-function relationships. By achieving these goals, the paper aims to provide an accurate view of protein functions across diverse human tissues. These issues form the foundation for the development of the proposed **OhmNet** model, which aims to address these limitations and provide a more accurate and biologically meaningful model for studying protein functions across human tissues.
Literature Review / Background Study	The literature surrounding gene function prediction and prioritization of disease-related genes has evolved significantly, integrating various computational methods and biological insights. Below is a brief overview of key contributions from the referenced papers: 1. **eneTIER: Prioritization of Candidate Disease Genes Using Tissue-Specific Gene Expression Profiles (Antanaviciute, A. et al., 2015)** This paper presents GeneTIER, a tool designed to prioritize disease-related genes based on tissue-specific gene expression data. It aims to enhance the accuracy of gene-disease association predictions by considering the specific expression patterns of genes across different tissues. 2. **Gene Ontology: Tool for the Unification of Biology (Ashburner, M. et al., 2000)** This foundational paper discusses the Gene Ontology (GO), a bioinformatics initiative that provides a unified vocabulary for annotating genes and gene products. GO categorizes genes into three domains: biological process, molecular function, and cellular component, enabling better data integration and interpretation in biology. 3. **Hierarchical Multi-Label Prediction of Gene Function (Barutcuoglu, Z., Schapire, R. E., and Troyanskaya, O. G., 2006)** This paper introduces a hierarchical multi-label approach for predicting gene function. The method builds a hierarchy of gene functions and applies machine learning to predict the functions of genes based on their features and relationships in the hierarchy. 4. **Laplacian Eigenmaps and Spectral Techniques for Embedding and Clustering (Belkin, M. and Niyogi, P., 2001)** This paper proposes the Laplacian eigenmaps algorithm, a spectral technique for dimensionality reduction and clustering. It is particularly useful for capturing the geometric structure of data and is applied to problems like image processing and gene expression analysis. 5. **Minimum Curvilinearity to Enhance Topological Prediction of Protein Interactions by Network Embedding (Cannistraci, C. V., Alanis-Lobato, G., and Ravasi, T., 2013)**

This paper focuses on improving protein interaction prediction by introducing a network embedding technique that minimizes curvilinearity. This method enhances the accuracy of predicting interactions by leveraging topological properties of protein interaction networks.

6. **Siri of the Cell: What Biology Could Learn from the iPhone**
   (Carvunis, A.-R. and Ideker, T., 2014)
   This paper draws parallels between the iPhone ecosystem and cellular networks, proposing that biological systems could learn from the modular, flexible, and interconnected nature of technological systems. It emphasizes the importance of systems biology in understanding cellular functions.
7. **BRENDA in 2015: Exciting Developments in Its 25th Year of Existence**
   (Chang, A., Schomburg, I., Placzek, S., et al., 2014)
   This paper provides an update on BRENDA, a comprehensive enzyme information system. It highlights the advancements and new features in the database as it celebrates its 25th anniversary, particularly in terms of enzyme functional data and annotations.
8. **The BioGRID Interaction Database: 2015 Update**
   (Chatr-Aryamontri, A. et al., 2015)
   The paper presents an updated version of BioGRID, a database containing curated biological interaction data. It highlights improvements in the database's coverage of protein-protein, genetic, and chemical interactions, as well as new tools for querying and analyzing interaction data.
9. **A Global Genetic Interaction Network Maps a Wiring Diagram of Cellular Function**
   (Costanzo, M. et al., 2016)
   This research maps a comprehensive genetic interaction network, providing insights into the complex wiring diagram of cellular function. The study uncovers how genes work together to maintain cellular processes and offers a global perspective on gene interactions.
10. **Navigability of Interconnected Networks Under Random Failures**
    (De Domenico, M. et al., 2014)
    This paper investigates the navigability of interconnected networks when subjected to random failures. It explores how the robustness and navigability of networked systems are impacted by disruptions and provides insights into the design of more resilient interconnected networks.

Project Implementation Method (less than 2500 characters)	1. Multi-Layer Network Construction **Task 1: Data Collection and Preprocessing** OhmNet works on multiple datasets, primarily protein-protein interaction (PPI) networks and tissue-specific gene expression data. The data is curated from sources Human Protein Reference Database (HPRD), BRENDA Tissue Ontology, Gene Ontology (GO), and GTEx dataset (Genotype-Tissue Expression). **1.1 Preprocessing:** Data Cleaning: Removing redundant. **1.2 Network Construction:** Creating tissue-specific PPI layers by mapping protein interactions to specific tissues.Hierarchical Organization structuring tissues into a tree-like hierarchy where broader categories (e.g., organ systems) encompass more specific subcategories (e.g., individual tissues). **Task 2: Building the Multi-Layer Graph Representation** Each layer in the network represents interactions occurring in a specific tissue. The graph structure consists of: 1. Intra-layer edges: Connections within the same tissue. 2. Inter-layer edges: Relationships between corresponding proteins across different tissues. 3. Hierarchy constraints: Edges based on tissue similarities. 4. Unsupervised Feature Learning **Task 3: Embedding Single-Layer Networks** **OhmNet:** employs a neural network-based embedding approach similar to Node2Vec. Random Walk Sampling: Captures local and global connectivity using Depth-First Search (DFS) and Breadth-First Search (BFS). Skip-Gram Optimization: Uses a Skip-Gram model with negative sampling to ensure proteins with similar network neighborhoods are embedded closely together. **Task 4: Learning Multi-Layer Features with Hierarchical Regularization** Enforcing tissue dependencies ensures that protein embeddings from related tissues exhibit consistent similarities, reflecting their biological connections. By leveraging hierarchical structure constraints, the relationships between parent and child tissues help guide the refinement of feature representations. This process improves the model's ability to capture tissue-specific characteristics and their interdependencies.

Task 5: Objective Function Design

Single-layer embedding objectives (local network structure learning).

Hierarchical dependency objectives (enforcing tissue-specific feature consistency).

Task 6: Training Strategy

Random Initialization: Initializes embeddings randomly.

Stochastic Gradient Descent (SGD): Uses negative sampling for faster training.

Iterative Updates: Adjusts embeddings layer by layer, refining features based on hierarchy constraints.

Task 7: Predicting Tissue-Specific Cellular Functions

OhmNet is evaluated on a multi-label classification task where proteins are assigned cellular function labels.

Evaluation metrics:

Area Under the ROC Curve (AUROC)

Precision-Recall (AUPRC)

Task 8: Transfer Learning for Function Prediction in Uncharacterized Tissues

Transfers functional knowledge from annotated tissues to unlabeled tissues.

Achieves 20.3% higher accuracy compared to non-hierarchical approaches.

Task 9: Visualizing Multi-Scale Tissue Networks

t-SNE-based visualization shows meaningful clustering of brain tissue-specific networks, validating hierarchical feature learning.

Benefits of the Project (less	**Benefits:** 1. **Better Understanding of Proteins:** The paper emphasizes that OhmNet enhances the understanding of protein functions in specific tissues, which is crucial for insights into biological processes and disease mechanisms. 2. **Improved Disease Diagnosis:** The ability of OhmNet to predict tissue-specific cellular functions can lead to earlier and more accurate disease diagnosis. 3. **Drug Development Support:** The paper discusses how insights from OhmNet can aid in identifying potential drug targets, facilitating the development of new treatments, including protein-based drugs.
Technical Details of Final Deliverable (less than 2500 characters)	1. OhmNet is a deep learning model capable of predicting protein functions. 2. A dataset of tissue-specific protein interactions for validation. 3. Visualization tools (Cytoscape) will use tissue-specific function mapping.

Final Deliverable of the Project	The final deliverable of this project would be to predict the protein functions in a multilayer tissue network. A deep learning based model OhmNet would be used to more accurately predict cellular function and also generates more accurate hypotheses about tissue-specific protein actions.
Core Industry (Optional)	As suggested by Supervisor
Other Industries (Optional)	
Core Technology	
Other Technologies (Optional)	
Sustainable Development Goals (Optional)	

References

1. Zitnik, M., & Leskovec, J. (2017). Predicting Multicellular Function through Multi-layer Tissue Networks. Bioinformatics, 33, i190-i198.

2. Greene, C. S., et al. (2015). Understanding Multicellular Function and Disease with Human Tissue-Specific Networks. Nature Genetics, 47(6), 569-576.

This proposal outlines a cutting-edge approach to leveraging AI in bioinformatics, with significant potential for real-world applications in disease research and personalized medicine.

Project Key Milestones

Elapsed time in (days or weeks or month or quarter) since start of the project	Milestone	Deliverable
Month 1-2	Data collection and preprocessing	Literature review document and data set compiled.
Month 3	Network construction using graphs	Architecture designed.
Month 4	Objection Design and Preliminary model development.	Initial model implemented.
Month 5-6	Model training strategies and validation	Trained model , performance report.
Month 7-8	Refinement and testing for improved accuracy	Optimized model , testing results.

Project Equipment Details

Item(s) Name	Type	No. of Units	Per Unit Cost (in Rs)	Total (in Rs)
GPU-enabled computing resources for model training.				
Database access for protein interaction networks.				
Python libraries (TensorFlow, PyTorch, NetworkX) for implementation.				
			Total in (Rs)	